

NSFDC Concessional Loan Digital Portal

System Architecture & Complete Project Specification

Government of India | Ministry of Social Justice & Empowerment | National Scheduled Castes Finance & Development Corporation

1. Executive Overview

The **NSFDC Concessional Loan Digital Portal** is an end-to-end, enterprise-grade digital platform developed for the **National Scheduled Castes Finance & Development Corporation (NSFDC)** under the *Ministry of Social Justice & Empowerment, Government of India*. The platform digitizes and automates the complete lifecycle of concessional credit allocation for Scheduled Caste (SC) beneficiaries across India—from multi-lingual voice/conversational intake, AI-driven OCR document verification, dynamic EMI amortization calculation, and PostGIS-powered spatial channel partner routing, to branch officer lead processing.

2. High-Level System Architecture

The system follows a containerized, decoupled client-server micro-services architecture orchestrated via Docker Compose:

- **Frontend Client Layer:** Next.js 14 App Router, React 18, TypeScript, Tailwind CSS, Leaflet GIS Maps, Web Speech API (Speech Recognition & Speech Synthesis).
- **API Gateway & REST Layer:** FastAPI (Python 0.110.0), Uvicorn ASGI server, Pydantic v2 validation schemas, CORS Middleware, Async Router.
- **Business Logic & Engine Core:** Financial Amortization Engine, GIS Spatial Router & Composite Score R_score Engine, PyTesseract OCR Verification Engine, Scheme Recommendation Engine.
- **Spatial Database Layer:** PostgreSQL 16 + PostGIS 3.4 (`postgis/postgis:16-3.4` container) with spatial indexing (GiST index on PostGIS `Point` geometry), custom ENUM types (`partner\_tier`, `app\_status`), 10 statutory schemes pre-loaded, and multi-state channel partner seeds.

3. Technology Stack & Tooling Specifications

Layer	Technology / Package	Version	Purpose / Responsibility
Frontend	Next.js / React	14.1.0 / 18.2.0	App Router, SSR, Client Components, State Management
Frontend	Tailwind CSS	3.4.1	Responsive Glassmorphic UI & Government Palette Styling
Frontend	Leaflet / React-Leaflet	1.9.4	Interactive GIS Mapping, Branch Markers & Spatial Polylines
Frontend	Web Speech API	Native Browser	Speech Recognition (STT) & Speech Synthesis (TTS) in 8 languages
Backend	FastAPI / Uvicorn	0.110.0 / 0.28.0	High-Performance Asynchronous Python REST Web API Server
Backend	Pydantic & Settings	2.6.4 / 2.2.1	Data Validation, Schema Enforcement & Env Variables
Backend	SQLAlchemy / GeoAlchemy2	2.0.28 / 0.14.7	ORM Data Mapping & Spatial PostGIS Queries
Backend	PyTesseract & Pillow	0.3.10 / 10.2.0	OCR Text Extraction & Image Sharpening/Preprocessing
Database	PostgreSQL + PostGIS	16-3.4	Relational & Spatial Database with GiST Indexing
Container	Docker Compose	3.8 Spec	Multi-container DB, Backend API & Frontend Web deployment

4. Deep-Dive into Business Logic Engines

4.1 Financial Amortization Engine (financial_engine.py)

Enforces the **Statutory Hard Gate**: Annual Family Income ≤ ₹5,00,000 INR. If exceeded, throws an HTTP 422 STATUTORY_INELIGIBILITY_INCOME_EXCEEDED exception. Formulates government concessional loan share and EMI calculations:

Principal Loan Amount: $P = \min(\text{Project Cost} \times \text{Govt Share \%}, \text{Max Scheme Cap})$

Beneficiary Margin Money: $M = \max(0, \text{Project Cost} - P)$

Monthly EMI Calculation: $EMI = P \times [r(1+r)^n / ((1+r)^n - 1)]$

where $r = (\text{Annual Interest Rate} / 100) / 12$ and $n = (\text{Tenure Years} \times 12) - \text{Moratorium Months}$.

4.2 GIS Spatial Router & Composite Scoring Engine (postgis_router.py)

Computes Haversine distances between beneficiary coordinates (lat, lon) and channel partner branches. Applies **Statutory Pruning Gates** to exclude branches where *Remaining Quota* ≤ 0 or *NPA %* ≥ 15.0%. Computes multi-factor composite ranking index **R_score**:

Composite Index: $R_score = (0.45 \times S_dist) + (0.35 \times S_quota) + (0.20 \times S_npa)$

Distance Sub-score: $S_{dist} = 1 / (1 + \text{Distance_km} / 10)$

Quota Utilization Sub-score: $S_{quota} = (\text{Allocated Quota} - \text{Utilized Quota}) / \text{Allocated Quota}$

NPA Health Sub-score: $S_{npa} = 1 - (\text{NPA \%} / 15.0)$

Status Pin Classification: GREEN ($R_score \geq 0.70$), YELLOW ($0.50 \leq R_score < 0.70$), RED_PRUNED (Excluded).

4.3 AI OCR Document Verification Engine (ocr.py)

Uses PyTesseract with PIL pre-processing (2.0x contrast enhancement, sharpening, greyscale). Validates SC Community Certificates using positive statutory keyword matching (SCHEDULED CASTE, COMMUNITY CERTIFICATE, TAHSILDAR, MEESEVA) while disqualifying non-eligible categories (OBC, SCHEDULED TRIBE, GENERAL CATEGORY).

5. Official Statutory NSFDC Schemes Directory (10 Schemes)

ID	Scheme Name	Category	Max Cost	Male %	Female %	Morat.	Tenure
NSFDC_MCF	Micro Credit Finance	MICRO	■1.4 Lakh	6.50%	5.50%	3 mos	3 yrs
NSFDC_MSJ	Mahila Samridhi Yojana	MICRO_WOMEN	■1.4 Lakh	N/A	5.00%	3 mos	3 yrs
NSFDC_TL	Term Loan General Scheme	TERM	■50.0 Lakh	7.50%	7.00%	6 mos	5 yrs
NSFDC_ELS_D	Educational Loan (Domestic)	EDU_DOMESTIC	■20.0 Lakh	7.00%	6.00%	12 mos	5 yrs
NSFDC_ELS_O	Educational Loan (Abroad)	EDU_ABROAD	■50.0 Lakh	7.50%	6.50%	12 mos	7 yrs
NSFDC_GBS	Green Business Scheme	GREEN_ENERGY	■30.0 Lakh	7.00%	6.50%	6 mos	5 yrs
NSFDC_LVY	Laghu Vyavsay Yojana	SMALL_BUSINESS	■5.0 Lakh	7.00%	6.50%	6 mos	4 yrs
NSFDC_SUY	Swachhta Udyami Yojana	SANITATION	■50.0 Lakh	6.00%	5.50%	6 mos	7 yrs
NSFDC_SSY	Shilpi Samridhi Yojana	ARTISAN	■1.4 Lakh	6.00%	5.00%	3 mos	3 yrs
NSFDC_MKY	Mahila Kisan Yojana	AGRI_WOMEN	■1.4 Lakh	N/A	5.00%	3 mos	3 yrs

6. Pan-India Channel Partner Network & Tiers

The portal integrates channel partners across all 28 States and UTs, structured into 3 distinct tiers:

- **State Channelizing Agencies (SCA):** State government welfare corporations (e.g., AP State SC Co-op Finance Corp, Telangana SC Co-operative Dev Corp, Mahatma Phule BC Dev Corp MH, TAHDCO TN, UP SC Finance & Dev Corp, DSFDC Delhi).

- **Public Sector Banks (PSB):** Nationalized commercial banks providing extensive branch reach (e.g., State Bank of India, Bank of Maharashtra).
- **Regional Rural Banks (RRB):** Micro and rural branch network (e.g., Chaitanya Godavari Grameena Bank, Telangana Grameena Bank, Karnataka Vikas Grameena Bank).

7. Accessibility: Multi-Lingual & Native Voice Intake Engine

To break down literacy barriers for rural SC beneficiaries, the frontend incorporates native localization for **8 Indian Languages** (English, Hindi, Bengali, Gujarati, Kannada, Marathi, Tamil, Telugu) and an interactive **Live Voice Assistant** leveraging Web Speech API for real-time Speech-to-Text (STT) form auto-filling and Text-to-Speech (TTS) audio guidance.

8. Project File & Codebase Structure

- **nsfcd-concessional-portal/docker-compose.yml:** Multi-container orchestration (PostGIS DB, FastAPI Backend, Next.js Frontend).
- **backend/app/main.py:** FastAPI entry point with CORS setup and OpenAPI documentation.
- **backend/app/core/constants.py:** Definitions for 10 NSFDC statutory schemes, income limits, and 28 Indian states.
- **backend/app/db/init_db.sql:** PostGIS schema, spatial extension, ENUM types, and seed data for schemes and channel partner branches.
- **backend/app/services/financial_engine.py:** Statutory income ceiling checks and loan amortization mathematics.
- **backend/app/services/postgis_router.py:** Haversine spatial calculations, statutory pruning gates, and R_score index calculations.
- **backend/app/api/v1/endpoints/ocr.py:** PyTesseract OCR document authentication and community keyword matching.
- **backend/app/api/v1/endpoints/routing.py:** Application lead dispatch, spatial branch locator, and tracking APIs.
- **frontend/src/app/apply/page.tsx:** Step-by-step intake wizard (voice/text, document upload, scheme match, GIS route dispatch).
- **frontend/src/app/channels/page.tsx:** Pan-India channel partner locator map with state filter and radius controls.
- **frontend/src/app/schemes/page.tsx:** Statutory scheme explorer and dynamic EMI calculation sliders.
- **frontend/src/app/track/page.tsx:** Real-time beneficiary application lifecycle tracking desk.
- **frontend/src/app/officer/page.tsx:** Channel partner branch officer lead desk for document inspection and sanction workflows.

9. End-to-End Application Lifecycle Flow

1. **Beneficiary Intake:** Applicant enters project cost, income, and activity purpose via conversational text or voice assistant.
2. **Statutory Income Gate:** Income checked against ₹5,00,000 ceiling. System matches beneficiary to optimal NSFDC scheme.
3. **OCR Document Verification:** Beneficiary uploads SC certificate. PyTesseract pre-processes and authenticates keywords in real-time.
4. **Amortization Computation:** EMI, moratorium period, government loan principal, and beneficiary margin calculated.
5. **PostGIS Spatial Routing:** System evaluates nearby branches, prunes invalid candidates, calculates R_score, and dispatches lead to top branch.
6. **Lifecycle Status Tracking:** Reference ID (e.g., SC-2026-AP9042) generated. Branch officer reviews lead and advances status: SUBMITTED → DOCS_VERIFIED → ROUTED_TO_CHANNEL → FIELD_INSPECTED → SANCTIONED → DISBURSED.

Document generated automatically for NSFDC Concessional Loan Digital Portal | Govt. of India